

Incubation time required for neonatal blood cultures to become positive

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Aim: We aimed to determine the laboratory detection time of bacteraemia in neonatal blood cultures, and whether this differed by: organism; samples deemed to represent true bacteraemia versus contaminants; and blood cultures collected from an infant <48 h of age (early) or ≥48 h of age (late).

Methods: A retrospective audit of all positive blood cultures collected from neonates in the Grantley Stable Neonatal Unit, Royal Women's Hospital, Brisbane, between 1 January 2000 and 31 December 2004 was undertaken. The bacteraemia detection method used was the BacTAlert system with Peds bottles.

Results: Two hundred and three positive blood cultures were included in the analysis. One hundred and sixteen (57%) were deemed septicaemia, 87 (43%) were deemed contaminants. The median (interquartile range) time to positivity for positive blood cultures deemed septicaemia and contaminants were 15.9 (11.6, 22.2) and 30.2 (20.4, 43.9) h, respectively. Fifty-six (28%) positive blood cultures were collected when infants were <48 h of age and 147 (72%) were collected in infants ≥48 h of age. *Post hoc* analysis revealed that the time to positivity for early septicaemia was 13.7 (11, 16.7) h; early contaminant was 25.2 (19.2, 33.8) h; late septicaemia was 17.2 (12.2, 23.4) h; and late contaminant was 37.9 (21.7, 51.2) h. The time to positivity for: Group B streptococcus was 9.3 (8.2, 11.0) h; *Escherichia coli* was 11.3 (10.0, 13.5) h; and coagulase-negative staphylococci was 28.9 (20.5, 41.2) h.

Conclusion: The incubation time for positive blood cultures significantly differs by organism type and whether they are considered early or late septicaemia versus contaminants. We recommend that: infants who are <48 h of age at the time of blood culture collection, who remain clinically well and have negative cultures 36 h after the initial collection can safely have their antibiotic treatment ceased; infants who are ≥48 h of age at the time of collection should continue antibiotic treatment for at least 48 h before cessation is considered.

Key words: bacteriological technique; blood culture; infant; newborn; septicaemia.

Neonatal sepsis is a potentially life-threatening event. The incidence of early-onset neonatal sepsis has been reported as less than 1–8.1 cases per 1000 live births.¹ Stoll *et al.* found the mortality rate of

neonatal septicaemia to be 38 deaths per 100 000 live births in the period 1992 through 1994.² Case reports of mortality rates for septic neonates range from 10% to 50%.^{1,3,4}

A positive blood culture is still considered the gold standard for the diagnosis of septicaemia.⁵ Positive results can be influenced by a number of factors including: collection technique, volume of blood drawn, number of colony-forming units present in the blood at the time of collection and the use of intrapartum antibiotics.^{5–7}

Most infants who have a blood culture collected are commenced on antibiotics while results are awaited. Our current practice is to treat the infant with intravenous antibiotics for a minimum 48 h after the blood culture has been collected.

The majority of babies that have blood cultures taken will not have any organisms detected and will not have septicaemia. Knowing the incubation time for the organisms that cause septicaemia will determine how long it is necessary to wait for a blood culture to return a positive result (along with other information) before declaring that the baby is unlikely to have septicaemia and does not require further antibiotics.

Prior to the routine use of automated blood culture incubators, Pichichero *et al.* reported that 96% and 98% of cultured pathogens

Key Points

- 1 Incubation time for positive blood cultures significantly differs by organism type.
- 2 Incubation time for positive blood cultures significantly differs whether or not they are considered early or late septicaemia versus contaminants.
- 3 Infants who are <48 hours of age at the time of blood culture collection, who remain clinically well and have negative cultures 36 hours after the initial collection can safely have their antibiotic treatment ceased.

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were positive at 48 and 72 h, respectively.⁸ This finding is, however, based on methods that have been superseded in modern microbiology laboratories. A number of authors have reported the duration of incubation required for detection of neonatal bacteraemia for various automated blood culture machines.^{9–15} Recommended durations of incubation have varied from 24 to 96 h.

A number of investigators have reported the time to positivity by organism type using an automated blood culture machine.^{10,12–15} However, the data are limited with regards to exact time to positivity, especially on key organisms such as Group B streptococci (*Streptococcus agalactiae*) (GBS).

Only one study separated groups into early onset (≤ 72 h of life) and late onset (> 72 h) and recommended that in asymptomatic infants in the early group it may be reasonable to shorten the length of antibiotic treatment to 24–36 h and in the late onset group shorten the length of treatment to 48 h.¹⁰

Few authors have attempted to determine the time to blood culture positivity according to whether the blood culture is deemed a contaminant or not.^{8,12,13} A number of authors have excluded suspected contaminants from their analysis.^{11,14}

We aimed to determine how long it took for positive blood cultures to be detected using an automated blood culture incubator. We also aimed to determine whether the time to detection differs by organism type and whether the organism is determined to represent true bacteraemia or a contaminant. Also, early (infant < 48 h of age) and late (infant ≥ 48 h of age) culture collections were analysed to determine whether there was any difference in time to positivity.

Methods

We retrospectively evaluated all positive blood cultures collected in the Grantley Stable Neonatal Unit Royal Brisbane and Women's Hospital, Brisbane, Queensland, Australia between 1 January 2000 and 31 December 2004.

Blood cultures are collected on all infants who have risk factors for septicaemia or clinical signs suggestive of infection.¹⁶ These are based on the CDC guidelines for the prevention of GBS disease.¹⁷ In our nursery the usual practice is to collect a single blood culture. Skin is prepared with chlorhexidine/alcohol solution or isopropyl alcohol wipe prior to collection. The minimum volume of blood required for collection is 0.5 mL with a maximum of 2 mL; however, actual volumes collected are not recorded.

During the study period, all blood samples were inoculated aseptically into PEDS BacT/Alert blood culture bottles (bioMérieux Inc, Hazelwood, USA). These bottles were placed in the BacT/Alert microbial detection machine (bioMérieux Inc). The machine continuously agitates the bottles at 35°C. Bacterial growth produces carbon dioxide, changing the colour sensor in the bottom of the blood culture bottle from green to yellow. A light-emitting diode projects onto the sensor and as more carbon dioxide is produced, increased light is reflected. Carbon dioxide levels are measured at 10-min intervals and compared with the initial level. Software enables determination of length of time to positivity for each bottle. Time to positivity is calculated from the time the bottle is placed in the machine (not when the bottle is inoculated) to the time the machine flags carbon dioxide detection. Unfortunately, time of collection was not recorded for a large proportion of positive blood cultures and we were unable to collect these data for analysis.

It should be noted that if the blood culture is sent to the laboratory between the hours of 0700 and 2300 (7 days a week) then the bottle is placed immediately into the BacT/Alert microbial detection machine. However, if sent to the laboratory outside these times, the bottle is not placed into the machine until 0700 the following day. These cultures are stored at room temperature until entry into the blood culture machine.

For division into analytic data groups the following definitions apply based on Australian and New Zealand Neonatal Network recommendations.¹⁸ Blood cultures were taken for the investigation of early septicaemia if collected from infants < 48 h of age and for late septicaemia if collected from infants ≥ 48 h of age. All positive blood cultures are determined to be either due to septicaemia or due to a contaminant after review by the Director of Neonatology of the Grantley Stable Neonatal Unit for data entry into the neonatal database. Positive blood cultures were deemed to be consistent with septicaemia or a contaminant after consideration of all available clinical and laboratory evidence including whether or not the attending clinician decided to intentionally treat with antibiotics active against the identified bacteria. Treatment for a period of time > 72 h with an appropriate antibiotic was considered treatment with intent and the infant was considered to have septicaemia.

Statistical analysis

All statistical tests were performed using GraphPad Prism version 4.00 for Windows (GraphPad Software, San Diego, CA, USA, <http://www.graphpad.com>). Non-parametric data medians were compared for statistically significant differences using either the Mann–Whitney test or the Kruskal–Wallis test and Dunn's multiple comparison test.

Results

During the study period there were 5817 admissions to our level 2 and 3 nurseries. A total of 3509 blood cultures were collected during this period. Seven hundred and forty-six blood cultures were collected on infants of less than 1000 g birthweight. A total of 2640 blood cultures were collected from infants < 48 h of life, and 869 blood cultures were collected from infants ≥ 48 h. Two hundred and eleven blood cultures from 193 patients returned a positive result (6%). Eight were not included in the final analysis because the exact time to positivity was not recorded. Three blood cultures had no time to positivity recorded at all. Three were recorded as ' < 24 ' h, one as ' > 24 ' h and one as '3 days'.

Of the 203 blood cultures that returned a positive result, 10 had two organisms detected. Only the first organism isolated has the time of detection recorded. Seven of these 10 samples were considered to be contaminants; three were thought to be consistent with polymicrobial septicaemia.

Of the 203 positive blood cultures included in the analysis, 116 (57%) were deemed to reflect septicaemia, and 87 (43%) were deemed contaminants. One of these contaminants was thought to be due to a contaminated blood culture bottle (this sample isolated a bacillus species and returned a time to positivity of 2 h). The median time to positivity for septicaemia and contaminants is shown in Table 1. The difference between the medians was 14.3 h and this difference is statistically significant (Mann–Whitney test, $P < 0.0001$).

Table 2 lists the results for the five most common organisms, or groups of organisms and Figure 1 shows the scatter plot for time to positivity for these groups. The median times to positivity for different organisms were significantly different (Kruskal–Wallis test, $P < 0.0001$).

Fifty-six positive blood cultures were collected when the infant was <48 h of age. The organisms identified and their interpretations

Table 1 Septicaemia versus contaminant, early versus late, early septicaemia versus early contaminant and late septicaemia versus late contaminant

Organism	n	Time to positivity (h)		
		Median	First quartile	Third quartile
Septicaemia	116	15.9	11.6	22.2
Contaminant	87	30.2	20.4	43.9
Early	56	19.2	13.7	28.6
Late	147	19.8	13.6	34.0
Early septicaemia	21	13.7	11.0	16.7
Early contaminant	35	25.2	19.2	33.8
Late septicaemia	95	17.2	12.2	23.4
Late contaminant	52	37.9	21.7	51.2

are listed in Table 3. One hundred and forty-seven blood cultures were collected in infants ≥ 48 h of age. The time to positivity for the early and late groups is shown in Table 1. The difference between the median time to positivity between the two groups was not significant (Mann–Whitney test, $P = 0.40$).

We performed a *post hoc* analysis on time to positivity by whether the positive blood cultures represented early contaminants, early septicaemia, late contaminants or late septicaemia (data are in Table 1 and scatter plots shown in Fig. 2). The median times to positivity were significantly different (Kruskal–Wallis test, $P < 0.0001$), with the differences between paired groups most significant between early septicaemia and early contaminant (Dunn's multiple comparison test, $P < 0.001$); and late septicaemia and late contaminant (Dunn's multiple comparison test, $P < 0.001$).

All organisms representing early septicaemia were detected by 24 h, except one organism, a *Corynebacterium* sp., which was positive at 92.2 h. That patient was clinically well and the full blood examination, collected at the time of blood culture collection, was considered normal: this almost certainly did not represent true septicaemia, however, was entered into the database as such because appropriate antibiotics were used for >72 h.

All organisms, except one, representing late septicaemia were detected by 48 h. This organism, *Enterococcus faecalis*, was detected at 89.3 h. The full blood examination in this infant revealed marked toxic change with left shift. The infant was clinically unwell and antibiotics were continued >48 h because of these clinical concerns.

Table 2 Data presented by organism type

Organism	n	Time to positivity (h)		
		Median	First quartile	Third quartile
Group B streptococcus	16	9.3	8.2	11.0
<i>Escherichia coli</i>	15	11.3	10.0	13.5
Coagulase-negative staphylococcus	88	28.9	20.5	41.2
<i>Staphylococcus aureus</i>	24	17.6	14.7	20.2
<i>Micrococcus</i> sp.	13	34.8	29.5	43.0

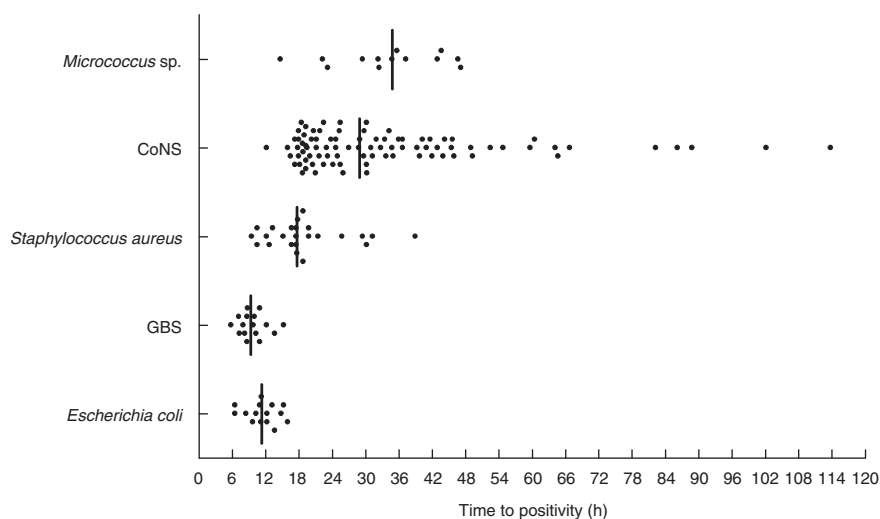
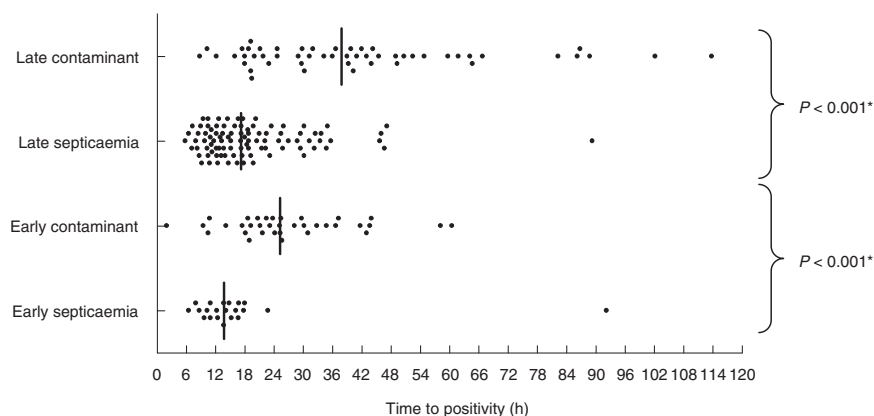


Fig. 1 Scatter plots showing time to positivity (h) for the five most common organisms or organism groups isolated in blood cultures during the study period. The vertical bar represents the median value. CoNS, Coagulase-negative staphylococci; GBS, Group B streptococcus.

Table 3 List of isolated organisms from early and late blood cultures, number and interpretation

Organism	Number	Septicaemia	Contaminant
Early cultures			
Coagulase-negative staphylococci	20	0	20
<i>Escherichia coli</i>	7	7	0
Group B streptococcus	5	5	0
<i>Micrococcus</i> sp.	4	0	4
<i>Staphylococcus aureus</i>	3	2	1
<i>Haemophilus influenzae</i>	3	3	0
<i>Bacillus</i> sp.	3	0	3
<i>Enterococcus faecalis</i>	3	0	3
<i>Pseudomonas</i> sp.	3	0	3
<i>Morganella morganii</i>	2	2	0
<i>Corynebacterium</i> sp.	1	1	0
<i>Salmonella</i> sp.	1	1	0
<i>Streptococcus acidominimus</i>	1	0	1
Total	56	21 (37.5%)	35 (72.5%)
Late cultures			
Coagulase-negative staphylococci	68	26	42
<i>Staphylococcus aureus</i>	21	18	3
Group B streptococcus	11	8	3
<i>Escherichia coli</i>	8	8	0
<i>Micrococcus</i> sp.	9	9	0
<i>Bacillus</i> sp.	4	4	0
<i>Enterococcus faecalis</i>	4	4	0
<i>Pseudomonas</i> sp.	4	3	1
<i>Enterobacter</i> sp.	4	4	0
<i>Klebsiella</i> sp.	4	4	0
<i>Acinetobacter</i> sp.	3	3	0
<i>Corynebacterium</i> sp.	2	0	2
<i>Salmonella</i> sp.	1	1	0
<i>Serratia marcescens</i>	2	2	0
<i>Candida parapsilosis</i>	1	0	1
<i>Pantoea agglomerans</i>	1	1	0
Total	147	95 (64.6%)	52 (35.4%)

Fig. 2 Scatter plot showing time to positivity (h) for blood cultures collected early (patient <48 h of age) and deemed a contaminant or septicaemia and blood cultures collected late (patient ≥48 h of age) and deemed a contaminant or septicaemia. The vertical bar represents the median value. *Dunn's multiple comparison test.



Discussion

Our study provides detailed analysis of time to positivity for blood cultures processed in an automated blood culture machine. In

particular, we have broken down our results into early and late groups of contaminant versus septicaemia. We have found statistically significant differences between the contaminant and septicaemia groups.

Including contaminants in our data analysis allows us to surmise that blood culture results returning a positive result after 24 h in the early group and 48 h in the late group are most likely to be contaminants as isolates considered to be contaminants took significantly longer to be detected. We chose to include the contaminated blood culture bottle in our results as this reflects the reality of clinical practice. Importantly, all blood cultures except for one in the early group were positive within the first 24 h of automated incubation. The single positive blood culture detected at 92.2 h in the early septicaemia group was a *Corynebacterium* sp. This was most likely a contaminant and the duration is outside our current clinical practice as antibiotic treatment is usually ceased at 48 h.

Our definition of septicaemia (in particular the continuation of antibiotic usage past 72 h) has probably overestimated the true incidence of septicaemia. Conversely, some infants who have septicaemia will return a negative blood culture. The numbers of samples in this category may have increased in recent years with the increasing use of intrapartum antibiotics.⁵ This study has not addressed this issue.

Our data revealed a higher incidence of septicaemia (vs. contaminant) in the late group (64.6%) compared with the early group (37.5%). This may be partially due to our definition of septicaemia (particularly antibiotic treatment >72 h). However, it may also reflect the population as an infant having a late culture is more likely to be premature and have vascular access devices that increase the risk of nosocomial bacteraemia.

Our results are similar to those of other authors who have analysed the results of blood cultures incubated in an automated blood culture machine.^{9,12–14} However, we have included the clinically relevant interpretation and analysis of all positive blood cultures including those thought to be contaminants. Compared with our study, Kumar *et al.* had a larger number of total positive blood cultures (451 positive cultures from 215 patients); they had a total of 2268 blood cultures.¹² They state that the median number of blood cultures from each baby is one; however, obviously more than one blood culture per baby must have been collected. It is unclear whether this collection occurred during the same episode of suspected septicaemia or on separate occasions. Only 30 of their positive blood cultures were collected in infants <48 h of age compared with our 56. We have classified some organisms as those consistent with septicaemia where previous authors would have deemed these to be contaminants (e.g. micrococcus).¹²

Garcia-Prats *et al.* recommended ceasing antibiotics in term asymptomatic babies undergoing sepsis work-up for maternal indications after 24–36 h.¹⁰ The major difference with our study is that we have separated our groups into babies undergoing early versus late sepsis work-up. Our implications for practice can be applied to both term and preterm infants. Our lab also uses a slightly different method of automated blood culture analysis (colour metric vs. headspace pressure). We recommend that this approach of ceasing antibiotics 36 h after collection only be applied to babies having their blood culture collected in their first 48 h of life.

We have identified that in the septicaemia group, the clinically important bacteria *Escherichia coli* and GBS were all detected by 16 h after entry into the automated blood culture machine. Pauli *et al.* found that the average time to detection for *E. coli* was 7.5 ± 0.7 h.¹⁴

We found that the median (interquartile range) time to positivity for *Staphylococcus aureus* was 17.6 (14.7–20.3) h and the median

(interquartile range) time to positivity for coagulase-negative staphylococci was 28.9 (20.5–41.2) h. This was similar to findings of other investigators.^{10,12,14}

Because of the possibility of a blood culture arriving at the laboratory between the hours of 2300 and 0700 and therefore the delayed entry into the blood culture machine, one either needs to include a safety margin of at least 8 h (plus transport time from site of collection to laboratory) into any interpretation, or needs to be able to access the exact time of entry into the automated blood culture machine before making a decision on cessation of antibiotics. We are now routinely recording these data as a result of this audit. We are also undertaking a separate study into the effects of delayed entry into the automated blood culture machine on the time to positivity.

By showing that a shorter duration of culture and therefore antibiotic treatment is unlikely to miss any clinically important infections, there is potential for significant cost savings in length of stay, antibiotics given and equipment used. Other major benefits are decreased admission time, enhanced mother/infant bonding and a reduction in the antibiotic pressure in the unit that inevitably leads to the emergence of antibiotic resistant bacteria colonising the patients and the environment.

This study is limited by it being a retrospective audit. Other limitations include the potential for classification bias as the director of Neonatology is responsible for the final decision as to whether a positive blood culture is deemed to be consistent with septicaemia or a contaminant. The relatively small number of positive blood cultures in the early group (56) and even smaller number deemed to be consistent with septicaemia (21) are of concern. However, we are reassured by the total number of blood cultures collected (2640) in this group and the time scale of 5 years.

Conclusion

The median (interquartile range) time to positivity for neonatal blood cultures using the BacT/Alert microbial detection system was 19.8 (13.7, 32.4) h. The length of time it takes for blood cultures to become positive does differ by organism type with *E. coli* and GBS growing fastest. There are statistically significant differences in time to positivity of blood cultures by whether the organism grown is considered to represent true septicaemia or a contaminant. There was no difference in median time to positivity by whether the blood culture is collected early (infant <48 h of age) or late (infant ≥48 h of age). All organisms representing early septicaemia grew in the blood culture machine by 24 h, except one organism that almost certainly was not a cause of a true septicaemia. All organisms representing late septicaemia, except one, were detected by 48 h. The exception was an infant that was clinically unwell and was continued on antibiotics.

Implications for clinical practice

We recommend that infants who are <48 h of age at time of blood culture collection, and who remain clinically well, cease antibiotic treatment 36 h after blood culture collection. Our data suggest that it may be safe to give the last dose of antibiotics 12 h after collection. However, we have allowed an extra 12 h (beyond 24 h required for growth of an early septicaemia organism) for possible delayed entry of the blood culture bottle into the automated blood culture

machine. This will decrease antibiotic duration and the associated nursing/medication/equipment costs by 12 h in those infants who remain asymptomatic. A significant proportion of term infants will be able to be discharged 12 h earlier to their mothers enhancing the mother/infant relationship. In decreasing the number of doses of antibiotics given, there is potential for incomplete treatment in those infants who have culture negative sepsis. It is unknown as to what minimum duration of antibiotics is required to adequately treat culture-negative septicaemia in an otherwise well infant.

We recommend continuing our current practice of allowing 48-h incubation time in infants >48 h of age at time of collection before considering the cessation of antibiotics (last dose of antibiotics at 48 h after collection). The reason for this difference in early versus late groups is predominantly due to the increased number of staphylococci that cause septicaemia in the late group and their increased median time to positivity (Table 2).

References

- 1 Pazzi D, Klein J, Baker C. Bacterial sepsis and meningitis. In: Remington JS, ed. *Infectious Diseases of the Fetus and Newborn Infant*, 6th edn. Philadelphia, PA: WB Saunders, 2006; 248–95.
- 2 Stoll BJ, Holman RC, Schuchat A. Decline in sepsis-associated neonatal and infant deaths in the United States, 1979 through 1994. *Pediatrics* 1998; **102**: e18.
- 3 Balaka B, Bonkougou B, Matey K, Napo-Bitanter S, Kessie K, Assimadi K. Neonatal septicaemia: bacteriological aspects and outcome in the University hospital center of Lome. *Bull. Soc. Pathol. Exotique* 2004; **97**: 97–9.
- 4 Tallur SS, Kasturi AV, Nadgir SD, Krishna BV. Clinico-bacteriological study of neonatal septicemia in Hubli. *Indian J. Pediatr.* 2000; **67**: 169–74.
- 5 Polin R, Parravicini E, Regan J, Taeusch H. Bacterial sepsis and meningitis. In: Taeusch W, Ballard RA, Gleason CA, eds. *Avery's Diseases of the Newborn*, 8th edn. Philadelphia, PA: WB Saunders, 2005; 551–7.
- 6 Neal PR, Kleiman MB, Reynolds JK, Allen SD, Lemons JA, Yu PL. Volume of blood, submitted for culture from neonates. *J. Clin. Microbiol.* 1986; **24**: 353–6.
- 7 Schelonka RL, Chai MK, Yoder BA, Hensley D, Brockett RM, Ascher DP. Volume of blood required to detect common neonatal pathogens. *J. Pediatr.* 1996; **129**: 275–8.
- 8 Pichichero ME, Todd JK. Detection of neonatal bacteremia. *J. Pediatr.* 1979; **94**: 958–60.
- 9 Escobar GJ, Zukin T, Usatin MS *et al.* Early discontinuation of antibiotic treatment in newborns admitted to rule out sepsis: a decision rule. *Pediatr. Infect. Dis. J.* 1994; **13**: 860–6.
- 10 Garcia-Prats JA, Cooper TR, Schneider VF, Stager CE, Hansen TN. Rapid detection of microorganisms in blood cultures of newborn infants utilizing an automated blood culture system. *Pediatrics* 2000; **105** (3 Pt 1): 523–7.
- 11 Hurst MK, Yoder BA. Detection of bacteremia in young infants: is 48 hours adequate? *Pediatr. Infect. Dis. J.* 1995; **14**: 711–13.
- 12 Kumar Y, Qunibi M, Neal TJ, Yoxall CW. Time to positivity of neonatal blood cultures. *Arch. Dis. Child.* 2001; **85**: F182–6.
- 13 Kurlat I, Stoll BJ, McGowan JE Jr. Time to positivity for detection of bacteremia in neonates. *J. Clin. Microbiol.* 1989; **27**: 1068–71.
- 14 Pauli I Jr, Shekhawat P, Kehl S, Sasidharan P. Early detection of bacteremia in the neonatal intensive care unit using the new BACTEC system. *J. Perinatol.* 1999; **19**: 127–31.
- 15 Rowley AH, Wald ER. Incubation period necessary to detect bacteremia in neonates. *Pediatr. Infect. Dis.* 1986; **5**: 590–1.
- 16 Davies MW, Cartwright DW. *Pocket Notes on Neonatology*. Brisbane: Mark Davies, 2003.
- 17 Prevention CfDCA. Prevention of perinatal group B streptococcal disease. *Morb. Mortal. Wkly Rep.* 2002; **51** (RR-11): 12–13.
- 18 Abeywardana S. *The Report of the Australian and New Zealand Neonatal Network, 2003*. Sydney: Australian and New Zealand Neonatal Network, 2005.